

Ink Density Saturation in Duan Inkstones

Anonymous submission

Stone	τ (min)	$D_{\max}$
Duan (Laokeng)	4.1	0.94
Duan (Majiakeng)	6.8	0.87
She	9.2	0.81

Table 1: Saturation parameters by stone.

Abstract

A minimal document in the AAAI 2027 submission format.
The content mirrors `latex-rendering-check.tex`:
ink density saturates as a function of grind time, and the
stone — not the stick — sets the ceiling.

Model

Let $D(t)$ denote ink density after t minutes of grinding.
Density follows a saturating exponential,

$$D(t) = D_{\max} \left(1 - e^{-t/\tau}\right), \quad (1)$$

where τ is the stone’s time constant and $D_{\max}$ its ceiling.
The relationship was established empirically by Tanaka and
Sato (2019).

Results

Table 1 lists time constants for three stones. The grinding
surface, not the ink stick, dominates.

Checklist

Opening this file and pressing preview should produce a
two-column PDF with the AAAI running head, a numbered
equation, a booktabs table, and a resolved citation to Tanaka
and Sato (2019) in the references.

References

Tanaka, Y.; and Sato, K. 2019. Ink Density Saturation Curves
for Traditional Duan Inkstones. *Journal of East Asian Ma-
terial Culture*, 12(3): 33–52.

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